Learning With Self-Calibrator for Fast and Robust Low-Light Image Enhancement

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***Abstract*—Convolutional Neural Networks (CNNs) have shown significant success in the low-light image enhancement task. However, most of existing works encounter challenges in balancing quality and efficiency simultaneously. This limitation hinders practical applicability in real-world scenarios and downstream vision tasks. To overcome these obstacles, we propose a Self-Calibrated Illumination (SCI) learning scheme, introducing a new perspective to boost the model's capability. Based on a weight-sharing illumination estimation process, we construct an embedded self-calibrator to accelerate stage-level convergence, yielding gains that utilize only a single basic block for inference, which drastically diminishes computation cost. Additionally, by introducing the additivity condition on the basic block, we acquire a reinforced version dubbed SCI++, which disentangles the relationship between the self-calibrator and illumination estimator, providing a more interpretable and effective learning paradigm with faster convergence and better stability. We assess the proposed enhancers on standard benchmarks and in-the-wild datasets, confirming that they can restore clean images from diverse scenes with higher quality and efficiency. The verification on different levels of low-light vision tasks shows our applicability against other methods.**

***Index Terms*—Low-light image enhancement, self-calibrator, additivity condition, low-light vision.**

I. Introduction

LOW-LIGHT image enhancement aims to reveal information hidden in the dark to improve image quality, and it has garnered significant attention in various emerging computer vision areas [1], [2], [3], [4], [5], [6], [7] recently. Although many advanced techniques have been developed in the past few years, how to construct an efficient and effective model for complex and diverse scenarios is still in suspense.

With the rapid advancements in deep learning, contemporary approaches to enhance low-light images are predominantly centered around the development of intricate deep models. A myriad of deep networks has surfaced, each adopting diverse architectural forms. The representative mechanism [8], [9], [10] focuses on creating decomposition-type architectures derived from fundamental physical knowledge. This approach involves breaking down low-light observations into illumination and reflectance components, corresponding to normal-light images. These architectures typically incorporate a dual-branch structure, introducing structural redundancy that may limit efficiency. Another category in architectural design attempts to move away from knowledge-driven patterns, instead heuristically constructing various structural forms [11], [12], [13]. While these specifically designed architectures often demonstrate excellent visual performance, they are usually time-consuming and less practical.

The exploration of learning paradigms has garnered significant attention across diverse fields [14], [15]. In the realm of low-light image enhancement, there has been a notable shift towards creating more efficient learning paradigms, with a specific emphasis on developing lightweight yet effective models. A straightforward approach involves defining various forms of training regularizations [16], [17], [18] to address the challenges associated with designing architectures. However, these efforts face limitations in terms of their visual quality to general scenarios due to the constraints imposed by the defined regularizations.

To settle the challenge of balancing efficiency and quality towards general scenarios, we develop a new Self-Calibrated Illumination (SCI)[1] learning framework for fast, flexible, and robust low-light image enhancement. By exploiting the intermediate output of the illumination learning, we construct a self-calibrated module to endow a stronger representation of the single basic block and convergence between the results of each stage to ensure efficiency. The specifically-designed unsupervised learning pattern further guarantees the generalization. More concretely, our main contributions can be concluded as:

- We develop an embedded self-calibrator for weight-sharing illumination learning to achieve stage-level convergence. This advancement improves exposure stability and

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The project about this work is publicly available at https://github.com/vis-opt-group/SCI.

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[1] A preliminary version of this work has been published in [1].

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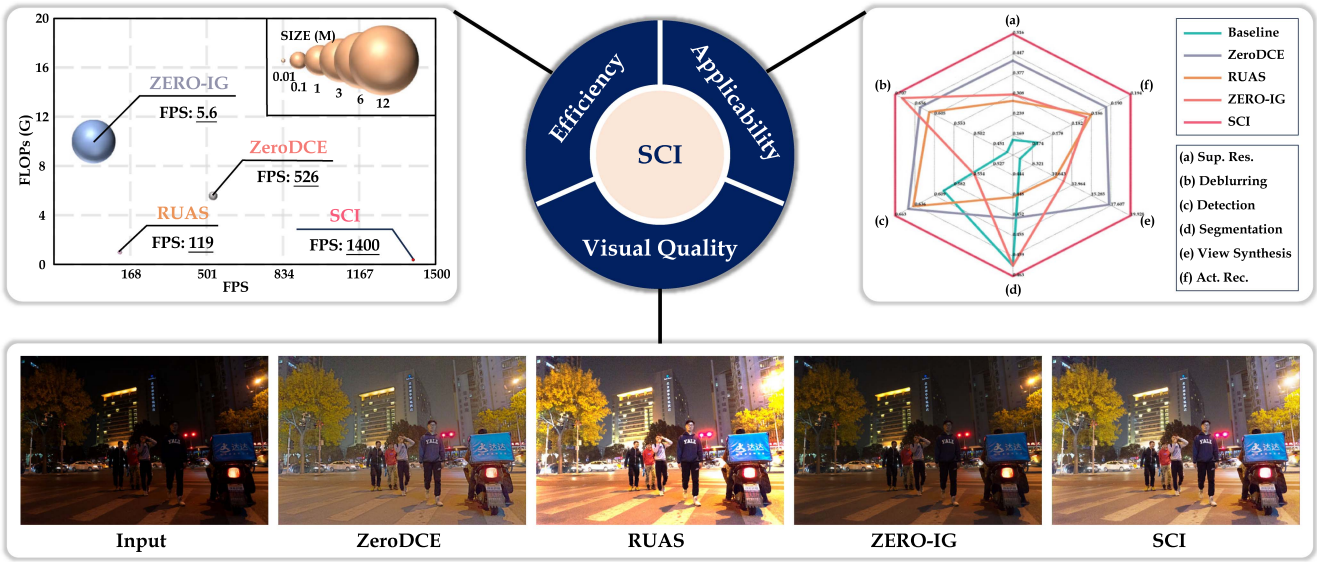


Fig. 1. Comparison among representative unsupervised methods (including ZeroDCE [16], RUAS [19], and ZERO-IG [20]) and our SCI. To provide a comprehensive evaluation, we report efficiency, applicability, and visual quality. Our method is remarkably superior to others from a statistical perspective regarding computational efficiency and applicability. As shown in the bottom part, these compared methods appear to have incorrect exposure, color distortion, and insufficient structure to degrade visual quality. In contrast, our result presents a vivid color and sharp outline. The abbreviation Sup. Res. and Act. Rec. in the top right part represents super-resolution and action recognition, respectively.

substantially alleviates the computational burden. To our knowledge, this represents the inaugural effort to expedite the deep low-light image enhancer by exploiting learning processes.

- By imposing the additivity condition on the fundamental block of the illumination estimator, we deduce a reinforced version (referred to as SCI++) featuring an independent self-calibrator. SCI++ resolves the intertwined dependencies in SCI's original architecture, establishing a transparent learning framework that achieves faster convergence and enhanced stability.
- We define the unsupervised training loss to control the output of each stage under the influence of the self-calibrator, thereby imparting a fast adaptive ability to diverse scenes. Property analysis reveals that our method exhibits operation-insensitive adaptability and model-irrelevant generality, characteristics not identified in existing works.
- Extensive experiments are performed to highlight our superiority. We verify various downstream tasks including denoising, super-resolution, deblurring, detection, segmentation, view synthesis, and action recognition, illustrating its applicability. In summary, this work rejuvenates low-light image enhancement encompassing advancements in visual quality, efficiency, and task applicability.

This work extends our preliminary version [1] and introduces several key improvements. First, we propose a reinforced version, SCI++, which untangles the entanglement between the estimator and the embedded self-calibrator. The newly introduced independent self-calibrator enhances transparency and interpretability in understanding the learning mechanism. Second, we make more comprehensive evaluations by testing more challenging examples and comparing recently-proposed algorithms, we also perform more analytical experiments. Lastly, we explore the potential of enhancers for three types of generalized downstream vision tasks including pixel-level (*e.g.,* deblurring), semantic-level (*e.g.,* segmentation), and other vision (*e.g.,* view synthesis), to further verify our effectiveness.

II. Related Work

In this section, we comprehensively introduce the existing related works from two aspects, including constructing network architectures and designing learning paradigms.

A. Constructing Network Architectures

Current deep learning models for low-light image enhancement emphasize diverse architectural constructions, incorporating both decomposition-type frameworks informed by physical principles and heuristic approaches.

In the realm of decompositional network design, the quintessential architecture for enhancing low-light images has evolved considerably, drawing inspiration from the foundational Retinex model. RetinexNet [8], a seminal work in Retinex-inspired deep networks, effectively estimated illumination and reflectance simultaneously. Building on this, subsequent studies had refined the modeling approaches for enhanced performance. The KinD [21] introduced stronger inter-component connectivity, with its subsequent version [9] further optimizing the architecture for faster training. Additionally, the approach [10] incorporated a sparse gradient minimization subnetwork to maintain key structural information during decomposition. Fan et al. [22] integrated a segmentation network, augmenting semantic reflectance understanding. Furthermore, the fusion of

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optimization-inspired unrolling techniques with deep networks has led to the creation of interpretable decomposition-type architectures, drawing considerable interest in the field [23], [24], [25], [26].

Apart from the aforementioned decomposition-type modeling mechanisms, a series of techniques have made efforts in heuristically designing architectures [11], [13], [27], [28]. Both the methods in [27] and [13] focused on building effective attention mechanisms to improve lightness recovery, while the work in [28] emphasized a pyramid embedding module for color correction. Wang et al. [11] designed a normalization flow to effectively characterize the structural context in the enhanced results. Yang et al. [29] proposed to exploit the controllable fitting capability of neural representation for underexposure image correction. Certain heuristic network architectures [12], [30], [31] enhance images by leveraging foundational models pre-trained on extensive datasets to acquire prior knowledge.

Very recently, low-light image enhancement has shifted from complex module design toward efficient architectures that maintain high performance. Zhang et al. showed that even a single-layer convolutional model can achieve comparable enhancement on CPUs [32]. Wang et al. leveraged frequency-domain priors via Fourier transforms to enhance detail in underexposed areas with minimal overhead [33], while Yan et al. exploited color-space conversion as a lightweight prior to design a compact, single-space model [34]. Fu et al. further balanced accuracy and efficiency for mobile use by combining streamlined convolutions with lightweight attention [35].

Instead of focusing on designing network architectures mentioned above, this work focuses on exploring the model learning process derived from physic-inspired optimization to realize fast and robust low-light image enhancement.

B. Designing Learning Paradigms

Recent research for low-light image enhancement has shifted its focus from architectural design to the development of learning strategies, influenced by advancements in learning methods across different fields [14], [15], [36], [37].

Wang et al. [38] designed a zero-reference learning framework by utilizing an illumination-invariant prior as the intermediary, realizing the ability without relying on any specific low-light data. Yang et al. introduce a efficient diffusion-based unsupervised enhancement method that leverages pretrained diffusion priors to improve generalization without costly supervision [39]. The work in [40] tried to build distribution discrepancy across different datasets by bilevel optimization to realize the fast adaption. The method in [17] formulated a multi-task joint contrastive learning framework, integrating high-level semantic knowledge with low-level signal priors. Addressing reflectance consistency across multi-exposure images, the approach proposed in [18] introduced a reference-free learning technique with adaptive constraints.

Mirroring the evolutionary trajectory of architectural design paradigms, formulating learning strategies that enable efficient model training has garnered increasing attention as a pivotal research direction. Guo et al. [16] introduced a lightweight unsupervised approach, intricately tailoring a set of loss functions to facilitate the learning of pixel-level high-order curves. Liu et al. [25] devised a cooperative learning framework, exploring the intrinsic connections between various visual tasks and low-light scene features. An et al. introduced auto re-reparameterization to convert multi-branch networks into a single-branch structure at inference, boosting efficiency without losing capacity [41]. Fu et al. propose a regularizer-free Retinex model trained via contrastive learning and self-distillation, enabling efficient and flexible low-light enhancement without explicit regularization [42].

Different from them, this work designs a simple but effective unsupervised learning paradigm with cross-scenario adaptability, it possesses the operation-insensitive adaptability and model-irrelevant generality that have been exploited in existing works.

III. SCI

In this section, we first introduce the illumination learning with weight sharing, then we build a self-calibrated module. Finally, we make a comprehensive discussion about our constructed SCI.

A. Weight-Sharing Illumination Estimator

According to the Retinex theory, there is a connection existing between the low-light observation $\mathbf{y}$ and the desired clear image $\mathbf{z}$: $\mathbf{y} = \mathbf{z} \otimes \mathbf{x}$, where $\mathbf{x}$ represents the illumination component. Generally, illumination is viewed as the core component that needs to be mainly optimized for low-light image enhancement. The enhanced output can be further acquired by removing the estimated illumination according to the Retinex theory. Here, inspired by the stage-wise optimization process for illumination presented in the works [19], [44], by introducing a mapping $\mathcal{H}_{\boldsymbol{\theta}}$ with parameters $\boldsymbol{\theta}$ to learn the illumination, we provide a progressive perspective to model this task, the basic unit is written as

$$\mathcal{F}(\mathbf{x}^t): \begin{cases} \mathbf{u}^t = \mathcal{H}_{\boldsymbol{\theta}}(\mathbf{x}^t), \mathbf{x}^0 = \mathbf{y}, \\ \mathbf{x}^{t+1} = \mathbf{x}^t + \mathbf{u}^t, \end{cases} \tag{1}$$

where $\mathbf{u}^t$ and $\mathbf{x}^t$ represent the residual term and illumination at tth stage ($t = 0, \ldots, T-1$), respectively. It should be noted that we do not mark the stage number in $\mathcal{H}_{\boldsymbol{\theta}}$ because we adopt the weight sharing mechanism, i.e., using the same architecture $\mathcal{H}$ and weights $\boldsymbol{\theta}$ in each stage.

In fact, the parameterized operator $\mathcal{H}_{\boldsymbol{\theta}}$ learns a simple residual representation $\mathbf{u}^t$ between the illumination and low-light observation. This process is inspired by a consensus, i.e., the illumination and low-light observation are similar or existing linear connections in most areas. Compared with adopting a direct mapping between the low-light observation and illumination (a commonly-used pattern in existing works, *e.g.,* [19], [45]), learning a residual representation substantially reduces the computational difficulty to both guarantee performance and improve the steadiness, especially for the exposure control.[2]

[2]Please refer to the ablation study in Section VII-E to confirm it.

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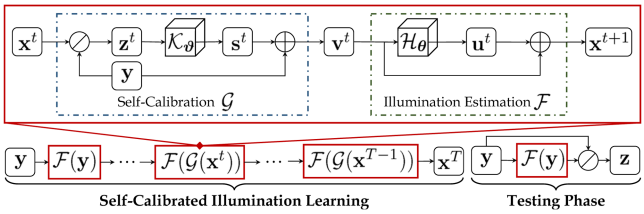


Fig. 2. The entire framework of SCI. In the training phase, our SCI is composed of the illumination estimation and embedded self-calibrator. The embedded self-calibrator map is added to the original low-light input as the input of the illumination estimation at the next stage. Note that these two modules are respectively shared parameters in the whole training procedure. In the testing phase, we just utilize a single illumination estimation module.

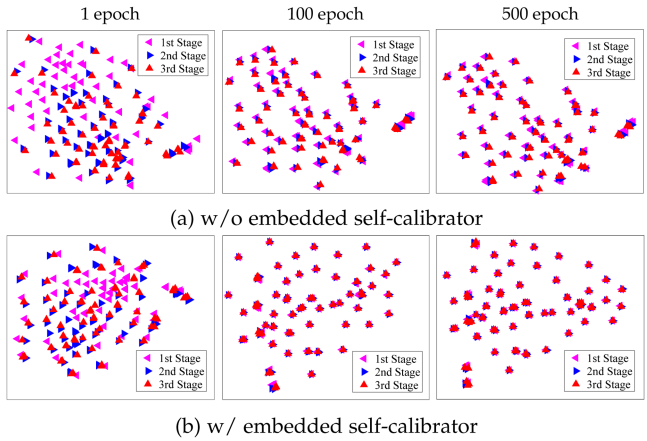


Fig. 3. Comparing t-SNE [43] distributions in terms of the results of each stage on whether using embedded self-calibrator. It exhibits why we can use a single stage for testing, that is, the results of each stage in SCI can rapidly converge to the same value, but w/o embedded self-calibrator cannot realize it all the time.

Indeed, we can directly utilize the above-built process with the given training loss and data to acquire the enhanced model. But it is noticeable that the cascaded mechanism with multiple weight sharing blocks inevitably gives a rise to foreseeable inference cost. *Revisiting this sharing process, each shared block expects to output a result that is close to the desired goal as far as possible. Going a step further, the ideal circumstance is that the first block can output the desired result, which satisfies task demands. Meanwhile, the latter blocks output the similar, even the completely same results as the first block does. In this way, in testing phase, we just need a single block to accelerate the inference speed.* Next, we will explore how to realize it.

B. Embedded Self-Calibrator

Here, we aim at defining a module to make results of each stage convergent to the same one state. We know that the input of each stage stems from the previous stage and the input of the first stage is definitely defined as the low-light observation. An intuitive idea is that whether we can bridge the input of each stage (except the first stage) and the low-light observation (i.e., the input of the first stage) to indirectly explore the convergence behavior between each stage. To this end, we introduce an embedded self-calibrator map $\mathbf{s}$ and add it to the low-light observation to present the difference between the input in each stage and the first stage. The embedded self-calibrator is presented as

$$\mathcal{G}(\mathbf{x}^t): \begin{cases} \mathbf{z}^t = \mathbf{y} \oslash \mathbf{x}^t, \\ \mathbf{s}_{\mathbf{e}}(\mathbf{x}^t) = \mathcal{K}_{\boldsymbol{\vartheta}}(\mathbf{z}^t), \\ \mathbf{v}^t = \mathbf{y} + \mathbf{s}_{\mathbf{e}}(\mathbf{x}^t), \end{cases} \tag{2}$$

where $t \geq 1$, $\mathbf{v}^t$ is the converted input for each stage, and $\mathcal{K}_{\boldsymbol{\vartheta}}$ is the introduced parameterized operator with the learnable parameters $\boldsymbol{\vartheta}$ for learning the self-calibration $\mathbf{s}_{\mathbf{e}}$. Then the conversion for the basic unit in tth stage can be written as $\mathcal{F}(\mathbf{x}^t) \rightarrow \mathcal{F}(\mathcal{G}(\mathbf{x}^t))$. Actually, our constructed embedded self-calibrator gradually corrects the input of each stage by integrating the physical principle to indirectly influence the output of each stage. To evaluate the effects of the embedded self-calibrator on the convergence, we plot tSNE distributions among results of each stage in Fig. 3, and we can easily observe that the results of each stage indeed converge to the same value. But this phenomenon cannot be found in the case without the embedded self-calibrator. Additionally, the above conclusion also reflects that we indeed accomplish the intention as described in the last paragraph of Section III-A, i.e., training multiple cascaded blocks with the weight sharing pattern but only using the single block for testing.

We also provide the overall flowchart in Fig. 2 for understanding our established SCI framework.

IV. SCI++

In this section, we first rethink SCI from a critical perspective. Then, we introduce an additivity condition into the illumination estimator. Finally, an independent self-calibrator is derived to build SCI++.

A. Rethinking SCI

In essence, the embedded self-calibrator plays an auxiliary role in learning a better basic block (the illumination estimation block in this work) that is cascaded to generate the overall illumination learning process with the weight sharing mechanism. More importantly, the embedded self-calibrator confers the convergence between results of each stage, it yet has not been explored in existing works. Moreover, the core idea of SCI is actually introducing the additional network module to assist in training, but not in the testing. It improves model characterization to realize that only using the single block for testing. That is to say, the mechanism "*weight sharing + task-related embedded self-calibrator*" may be transferred to handle other tasks for acceleration.

The SCI learning framework is typical to undergo multiple stages of learning but conduct testing with only one stage. This complexity arises from the intricate interplay between the illumination estimator and the embedded self-calibrator, making it challenging to provide a straightforward representation of the underlying cause. As depicted in the bottom part of Fig. 4, each illumination estimator is paired with a self-calibrator, except for

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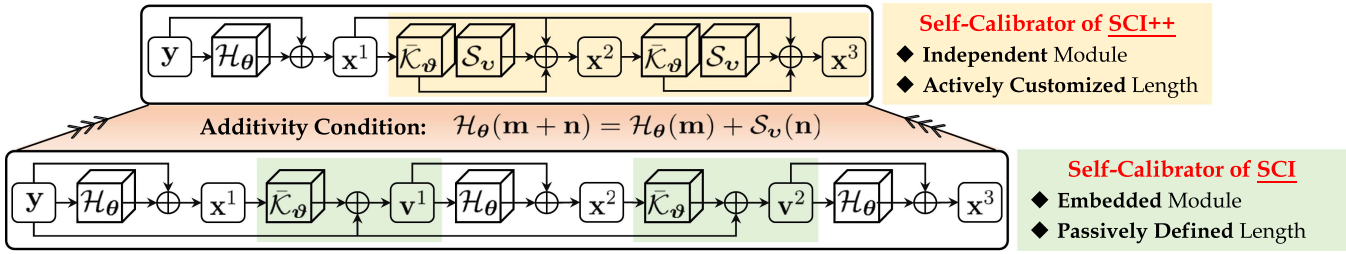


Fig. 4. Comparing the flowchart between the original method (i.e., SCI) and its reinforced version (i.e., SCI++). By enforcing the additivity condition into the illumination estimator, the self-calibrator undergoes a transition from being embedded to becoming independent and gains the advantage of active customized length for self-calibrator.

the first one. That is to say, the length of the embedded self-calibrator in SCI is passively defined according to the number of illumination estimators. This relationship can be expressed through a piecewise function. Specifically, when $t = 1$, we have

$$\mathbf{x}^1 = \mathbf{y} + \mathcal{H}_{\boldsymbol{\theta}}(\mathbf{y}), \tag{3}$$

otherwise when $t > 1$, we have

$$\mathbf{x}^t = \mathbf{y} + \mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1}) + \mathcal{H}_{\boldsymbol{\theta}}\left(\mathbf{y} + \mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1})\right), \tag{4}$$

where $\mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1}) = \bar{\mathcal{K}}_{\boldsymbol{\vartheta}}(\mathbf{x}^{t-1})$ denotes a two-step computational process, i.e.,

$$\bar{\mathcal{K}}_{\boldsymbol{\vartheta}}(\mathbf{x}^{t-1}) : \begin{cases} \mathbf{z}^{t-1} = \mathbf{y} \oslash \mathbf{x}^t, \\ \mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1}) = \mathcal{K}_{\boldsymbol{\vartheta}}(\mathbf{z}^{t-1}). \end{cases} \tag{5}$$

B. Introducing Additivity Condition

Combining the testing phase (as shown in the lower-right section of Fig. 2), we can observe that (3) is the crucial fundamental computational flow presented in both the training and testing phases. In other words, if we can delete the identical component with shared parameters $\mathcal{H}_{\boldsymbol{\theta}}$ in the latter phases ($t > 1$), the overall learning process will become more transparent and comprehensible.

To break up the entanglement between illumination estimator and embedded self-calibrator, we introduce the additivity condition,[3] described as

$$\mathcal{H}_{\boldsymbol{\theta}}(\mathbf{m}+\mathbf{n}) = \mathcal{H}_{\boldsymbol{\theta}}(\mathbf{m}) + \mathcal{S}_{\boldsymbol{\upsilon}}(\mathbf{n}), \tag{6}$$

where $\mathcal{S}_{\boldsymbol{\upsilon}}$ represents a newly-defined mapping that shares the same structure as $\mathcal{H}_{\boldsymbol{\theta}}$ but has different parameters. Additionally, $\mathbf{m}$ and $\mathbf{n}$ are two matrix-form variables.

By applying this condition to (4), we can derive the following modified process

$$\begin{aligned} \mathbf{x}^t &= \mathbf{v}^{t-1} + \mathcal{H}_{\boldsymbol{\theta}}(\mathbf{v}^{t-1}), \\ &= \mathbf{y} + \mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1}) + \mathcal{H}_{\boldsymbol{\theta}}(\mathbf{y} + \mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1})), \\ &= \mathbf{y} + \mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1}) + \mathcal{H}_{\boldsymbol{\theta}}(\mathbf{y}) + \mathcal{S}_{\boldsymbol{\upsilon}}(\mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1})), \\ &= (\mathbf{y} + \mathcal{H}_{\boldsymbol{\theta}}(\mathbf{y})) + \mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1}) + \mathcal{S}_{\boldsymbol{\upsilon}}\left(\mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1})\right), \\ &= \mathbf{x}^1 + \mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1}) + \mathcal{S}_{\boldsymbol{\upsilon}}\left(\mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1})\right). \end{aligned} \tag{7}$$

[3]Notice that the condition always satisfies and just be used for derivation, we ignore verifying it.

Based on the additivity condition presented in (6), we successfully transform (4) into an explicit form directly linked to the illumination generated in the initial stage, simplifying the computational flow in (4).

C. Independent Self-Calibrator

According to (7), we can build a new framework with independent self-calibrator, described as

$$\mathbf{x}^t = \mathbf{x}^1 + \mathbf{s}_{\mathtt{i}}(\mathbf{x}^{t-1}), \tag{8}$$

where $\mathbf{s}_{\mathtt{i}}(\mathbf{x}^{t-1}) = \mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1}) + \mathcal{S}_{\boldsymbol{\upsilon}}(\mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1}))$ represents the independent self-calibrator which integrates the original embedded self-calibrator.

In contrast to the embedded self-calibrator in the original SCI, the above formulation introduces an independent self-calibrator consisting of two sub-networks. In Fig. 4, the transition from SCI to SCI++ is illustrated. It is evident that the original self-calibrator in SCI exhibits a dependency on the illumination estimator, i.e., the self-calibrator is embedded in the cascaded illumination estimator, and its length is passively defined and determined by the length of the illumination estimator. Fortunately, by applying the additivity condition in the illumination estimator $\mathcal{H}$, we successfully break the entanglement, creating a new independent self-calibrator whose length is actively customized and irrelevant to the length of the illumination estimator.

V. Training Strategy

In this section, we introduce the unsupervised training loss for SCI and alternative learning for SCI++.

A. Unsupervised Training Loss for SCI

Considering the inaccuracy of existing paired data, we adopt the unsupervised learning to enlarge the network capability. We define the total loss $\mathcal{L}_{to}$ as

$$\mathcal{L}_{to} = \sum_{t=1}^{T} \mathcal{L}_u^t = \sum_{t=1}^{T} \left(\alpha \mathcal{L}_f(\mathbf{x}^t, \mathbf{s}_{\mathtt{e}}(\mathbf{x}^{t-1}); \mathbf{y}) + \beta \mathcal{L}_s(\mathbf{x}^t)\right), \tag{9}$$

where $\mathcal{L}_u^t$ is the loss function in tth stage, and T is the total stage number. The loss $\mathcal{L}_f$ and $\mathcal{L}_s$ represent the fidelity and smoothing loss, respectively. The parameters α and β are two positive balancing constants.

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Fidelity Loss: Its goal is to guarantee the pixel-level consistency between the estimated illumination and the input of each stage, formulated as

$$\mathcal{L}_f(\mathbf{x}^t, \mathbf{s}_\mathbf{e}(\mathbf{x}^{t-1}); \mathbf{y}) = \|\mathbf{x}^t - (\mathbf{y} + \mathbf{s}_\mathbf{e}(\mathbf{x}^{t-1}))\|^2. \tag{10}$$

Smoothness Loss: Constraining the illumination to satisfy the smoothness property is a broad consensus in this task [16], [21]. Here we adopt a smoothness term with spatially-variant ℓ_1 norm [47], presented as

$$\mathcal{L}_s(\mathbf{x}^t) = \sum_{i=1}^{N} \sum_{j \in \mathcal{N}(i)} w(i,j)|\mathbf{x}_i^t - \mathbf{x}_j^t|, \tag{11}$$

where N is the total number of pixels. i is the ith pixel. $\mathcal{N}(i)$ denotes the adjacent pixels of i in its 5×5 window. $w(i,j)$ represents the weight, whose formulated form is $w(i,j) = \exp(-\frac{\sum_c((\mathbf{y}(i,c)+\mathbf{s}_\mathbf{e}(i,c))-(\mathbf{y}(j,c)+\mathbf{s}_\mathbf{e}(j,c)))^2}{2\sigma^2})$, where c denotes image channel in the YUV color space [47]. $\sigma = 0.1$ is the standard deviations for the Gaussian kernels.

In fact, our defined training loss differs slightly from existing methods, with the primary distinction lying in the fidelity loss. Typically, the commonly-used form for measuring fidelity between illumination and low-light observation is expressed as $|\mathbf{x}^t - \mathbf{y}|$ in existing works [19], [40]. We know that the smoothness loss which operates in the image gradient space and the fidelity which functions in the image space. Based on this, a notable issue arises: *the smoothness has limited influence on the fidelity, which causes the illumination to closely resemble the low-light input, leading to overexposure in the enhanced result. This also explains why many illumination-based methods [8], [21], [44] require illumination adjustment after calculating the illumination.* Unlike these approaches, our method provides an explicit form to prevent this issue by monitoring the exposure state of intermediate output.

B. Alternative Learning for SCI++

Benefiting from the disentanglement between the illumination estimator and self-calibrator in SCI++, we can treat these two parts as separate but related sub-optimizations. Here we propose an alternative learning strategy by alternatively optimizing the estimator and self-calibrator. It can be described as the following computational process

$$\begin{cases} \boldsymbol{\theta}^{k+1} = \boldsymbol{\theta}^k - \eta_{\boldsymbol{\theta}}^a \frac{\partial \mathcal{L}_{to}^a(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} - \eta_{\boldsymbol{\theta}}^b \frac{\partial \mathcal{L}_{to}^b(\boldsymbol{\theta}, \boldsymbol{\pi}^k)}{\partial \boldsymbol{\theta}}, \\ \boldsymbol{\pi}^{k+1} = \boldsymbol{\pi}^k - \eta_{\boldsymbol{\pi}} \frac{\partial \mathcal{L}_{to}^b(\boldsymbol{\theta}^{k+1}, \boldsymbol{\pi})}{\partial \boldsymbol{\pi}}, \end{cases} \tag{12}$$

where $\boldsymbol{\pi} = \{\boldsymbol{\vartheta}, \boldsymbol{\upsilon}\}$ is the parameters of the independent self-calibrator, k represents the iterations in the training process. The above loss functions can be formulated as

$$\begin{cases} \mathcal{L}_{to}^a = \alpha_a \|\mathbf{x}^1 - \mathbf{y}\|^2 + \beta_a \mathcal{L}_s(\mathbf{x}^1), \\ \mathcal{L}_{to}^b = \alpha_b \|\mathbf{x}^T - (\mathbf{x}^1 + \mathbf{s}_\mathbf{i}(\mathbf{x}^{T-1}))\|^2 + \beta_b \mathcal{L}_s(\mathbf{x}^T). \end{cases} \tag{13}$$

Fig. 3(a) reveals that a multi-stage illumination estimator demonstrates a stage-level convergence tendency. SCI with an embedded self-calibrator seeks to diminish the domain gap between different stages to amplify this convergence tendency. In contrast, SCI++ offers an alternative perspective by directly optimizing the single-stage illumination estimator, thus avoiding the exploration of multi-stage cases. Essentially, the role of this independent self-calibrator is akin to that of a discriminator in GAN [48], aiding in improving the learning of "generator" which corresponds to the illumination estimator $\mathcal{H}$ in SCI++.

VI. Exploring Algorithmic Properties

In this part, we perform explorations on our proposed method to deeply analyze its algorithmic properties.

A. Operation-Insensitive Adaptability

In general, the operations used in network-based methods should be fixed and cannot be changed arbitrarily since these operations are acquired under the support of massive experiments. Fortunately, our proposed algorithm emerges the surprising adaptability on different exceedingly simple, even naive settings for $\mathcal{H}_{\boldsymbol{\theta}}$. In Table I, we can easily observe that our SCI acquire a stable performance among different settings (numbers of the block 3×3 convolution+ReLU). Further, we provide the visual comparison in Fig. 5, it can be easily observed that our method with different settings all brighten the low-light observation, showing very similar enhanced results. Revisiting our designed framework, this property can be acquired lies in SCI not only converts the consensus for illumination (i.e., residual learning) but also integrates the physical principle (i.e., element-wise division operation). This experiment also verifies the effectiveness and correctness of our designed SCI.

B. Model-Irrelevant Generality

Our proposed framework is actually a generalized learning paradigm if not limiting the task-related embedded self-calibrator, so ideally, it can be directly applied to existing works. Here, we take the recently-proposed representative work RUAS [19] as an example to make an exploration. Table II and Fig. 6 demonstrate the quantitative and qualitative comparison before/after using our SCI to train RUAS. Obviously, although we just utilized a single block (i.e., RUAS (1)) used in the unrolling process of RUAS to evaluate our training process, the performance still attains significant improvement. Specially, SCI++ performs better than SCI in the numerical scores. More importantly, our method can remarkably suppress the overexposure that appeared in the original RUAS. This experiment reflects our learning framework is indeed flexible enough and has a strong model-irrelevant generality. Moreover, it indicates that perhaps our method can be transferred to arbitrary illumination-based low-light image enhancement works.

C. Zero-Shot Learning Ability

Zero-shot learning [49] refers to the model's capability to generalize and perform well on new, unseen tasks without requiring additional training data. It is critical for real-world applications where models may encounter conditions or data distributions that differ significantly from the training data. Thanks to the unsupervised learning mechanism, here we try to explore it.

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TABLE I
QUANTITATIVE COMPARISON AMONG DIFFERENT SETTINGS FOR $\mathcal{H}_\theta$ ON MIT AND LSRW DATASETS

Setting for $\mathcal{H}_\theta$	Image Quality (*Left: MIT, Right: LSRW*)								Computational Efficiency	
Blocks: Channels	PSNR↑	SSIM↑	DE↑	NIQE↓	PSNR↑	SSIM↑	DE↑	NIQE↓	FLOPs (G)	TIME (S)
1: 3-3	24.930/**26.207**	0.932/0.944	7.254/7.318	3.011/3.156	18.224/18.728	0.609/**0.615**	6.968/6.924	3.568/3.592	**0.0202**	**0.0015**
2: 3-3-3	20.843/26.197	0.855/**0.948**	6.862/7.332	3.045/2.920	19.017/**19.902**	0.619/0.608	6.963/7.057	**3.563**/3.537	0.0410	0.0016
3: 3-3-3-3	**25.683**/25.826	**0.943**/**0.948**	7.268/**7.399**	**2.930**/2.924	**19.789**/19.835	0.611/0.597	**6.993**/**7.163**	3.616/**3.493**	0.0619	0.0017
3: 3-8-8-3	25.147/25.890	0.937/0.946	**7.297**/7.391	3.015/**2.913**	19.094/19.772	0.633/**0.611**	6.714/6.993	3.723/3.603	0.2503	0.0018
3: 3-16-16-3	19.029/26.011	0.876/0.947	6.909/7.383	3.095/2.926	19.742/19.323	**0.634**/0.611	6.851/6.982	3.629/3.636	0.7764	0.0022

"Blocks" and "Channels" represent the numbers of the basic block and the variation of channels in the basic block, respectively.



Fig. 5. Visual comparison among different cases in Table I. The low-light image comes from the MIT dataset.

TABLE II
QUANTITATIVE RESULTS OF MODEL-IRRELEVANT GENERALITY

Model	MIT Benchmark				LSRW Benchmark				Computational Efficiency	
	PSNR↑	SSIM↑	EME↑	NIQE↓	PSNR↑	SSIM↑	EME↑	NIQE↓	SIZE (M)↓	FLOPs (G)↓
RUAS (3)	19.198	0.912	**15.550**	**2.959**	15.828	**0.648**	11.570	4.993	0.0014	0.2831
RUAS (1) + SCI	24.937	0.928	15.435	3.134	19.033	0.621	20.216	3.439	**0.0005**	**0.0936**
RUAS (1) + SCI++	**25.714**	**0.934**	15.123	3.108	**19.495**	0.619	**20.363**	**3.420**	**0.0005**	**0.0936**

In which RUAS (d) represents adopting d iterative blocks for the unrolling process appeared in RUAS [19]. Here we adopt the LSRW [46] dataset for testing.

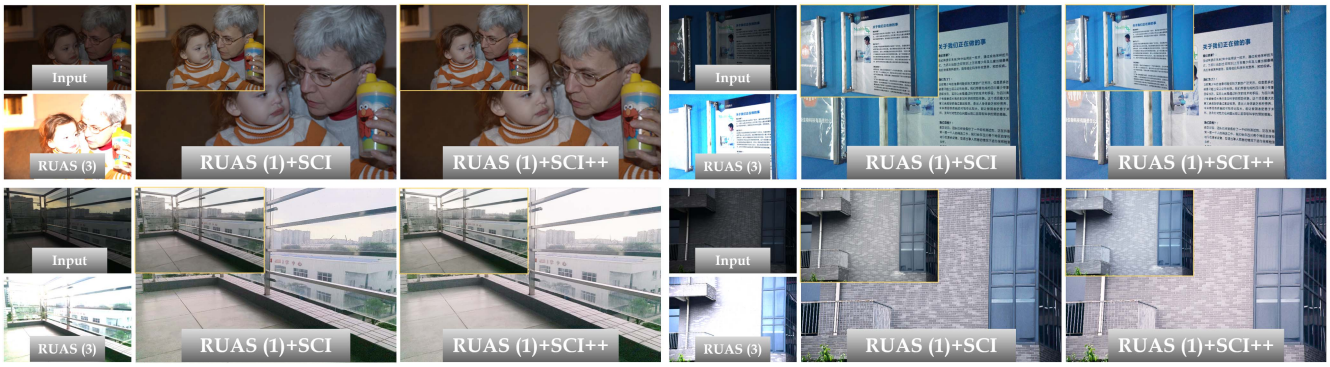


Fig. 6. Visual comparison among different cases in Table II. The samples of top and bottom come from the MIT and LSRW datasets, respectively.

As shown in Fig. 7(a), it was observed that the SCI exhibited significant oscillations during the early stages of training and underwent a sudden shift around 2×10^3 iterations before gradually stabilizing. In contrast, SCI++ showed a much smoother and more stable upward trend throughout the training process, with almost no major fluctuations. This indicates that SCI++ has better convergence and maintains higher stability during learning. As for visual effects, SCI++ produces superior results compared to the original SCI model as shown in Fig. 7(b). SCI++ exhibits a more refined capability to restore details, maintain color consistency, and reduce artifacts, which are the key factors. This experiment further reveals the latent application value of our method.

VII. EXPERIMENTAL RESULTS

In this section, we first provided all implementation details. Then we made experimental evaluations[4] on standard benchmarks and in-the-wild scenarios. Next, we evaluated computational efficiency. Finally, we explored the algorithm analyses from different aspects.

A. Implementation Details

Parameter Settings: In the training process, we used the ADAM optimizer [50] with parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$,

[4]All models for quantitative comparisons can be found on the project page.

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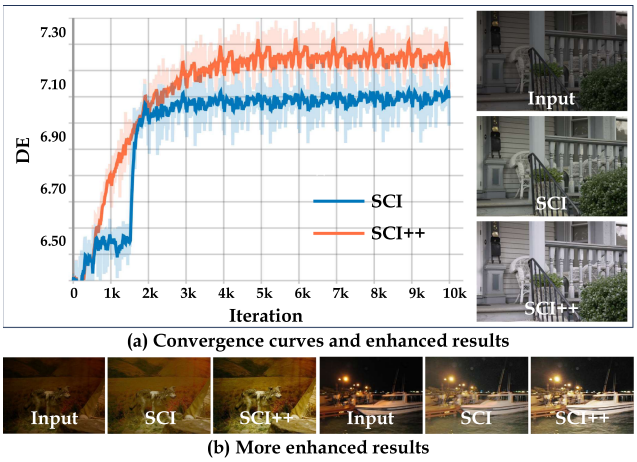


Fig. 7. Comparing convergence behaviors for zero-shot learning.

and $\epsilon = 10^{-8}$. The minibatch size was set to 8. The learning rate was initialized to 10^{-4}. The training epoch number was set to 1000. We adopted 3 convolution + ReLU with 3 channels as our default setting for $\mathcal{H}_{\boldsymbol{\theta}}$ in our all experiments according to the conclusion in Table I. The self-calibrated module contains four residual convolution layers, which ensures the lightweight of the training process. The network structure of self-calibrator may not be fixed, and we had done experiments to verify it in Section VII-E2.

Compared Methods: We compared our method with twelve advanced methods including RetinexNet [8], FIDE [51], KinD [21], DCC [28], URetinex [24], PairLIE [18], EnGAN [52], ZeroDCE [16], RUAS [19], CLIP-LIT [31], ZERO-IG [20], and QuadPrior [38].

Benchmarks Description and Metrics: To perform a full evaluation, we considered standard benchmarks and in-the-wild scenarios. The MIT dataset [53] (500 for training and 200 for testing) whose reference images were expertly corrected by professionals, and the LSRW dataset [46] (2000 for training and 200 for testing) whose inputs and references were captured using different exposure times, were used as the standard benchmarks. To further evaluate the algorithmic effectiveness in challenging scenarios, we constructed a new compositional In-the-Wild dataset[5] (2000 for training and 400 for testing) by randomly sampling from three widely-used real-world datasets: DARK FACE [54] (600 for training and 100 for testing), ExDark [55] (1000 for training and 200 for testing), and UFDD [56] (400 for training and 100 for testing). In terms of quantitative evaluation, we adopted three full-reference metrics including PSNR, SSIM and LPIPS, four no-reference metrics including DE [57], EME [58], LOE [59] and NIQE [59].

B. Evaluation on Standard Benchmarks

Quantitative Evaluation: As shown in Table III, our SCI model delivered competitive performance across multiple metrics, both excelling in full-reference and no-reference quality assessments. No-reference metrics, in particular, are crucial for image enhancement tasks where ground truth images are often unavailable, making these metrics more relevant for evaluating real-world performance. The strong performance in these no-reference metrics highlights SCI's ability to generate perceptually pleasing images under challenging conditions. In terms of full-reference metrics, SCI also performed on par with or better than many advanced deep networks, highlighting its ability to reconstruct images. Additionally, the SCI++ further achieved even higher scores across both reference and no-reference metrics. This improvement showcases the reinforced learning capacity of SCI++, affirming its ability to minimize perceptual distortions while maintaining accurate image features in quantitative evaluations.

Qualitative Evaluation: The qualitative analysis, as depicted in Figs. 8–9, further highlights the superiority of our models. Advanced deep learning networks (*e.g.,* URetinex, CLIP-LIT) always generated undesirable artifacts in the form of unknown veils, which obscured finer details and led to unnatural color distributions. These artifacts are particularly problematic in low-light enhancement tasks, where accurate color reproduction and detail preservation are essential. By comparison, our model consistently produced the highest visual quality, with vivid colors and prominent textures. The images enhanced by SCI and SCI++ appeared more natural and visually pleasing, with no washed-out effects or color inconsistencies seen in competing methods (*e.g.,* PairLIE, DCC). In particular, our two versions excelled in maintaining the integrity of fine details, producing sharper textures and clearer edges, which were often lost in other approaches. This is especially evident in challenging scenes with complex lighting conditions, where ours effectively balanced brightness, contrast, and detail restoration.

C. Evaluation on In-the-Wild Scenarios

To assess the robustness and generalization capability of our model in real-world conditions, we conducted evaluations using a dataset of 400 in-the-wild images collected from diverse and challenging environments. These images encompass a variety of scenarios, including low-light conditions, motion blur, high noise levels, and complex lighting, which represent practical situations the model may encounter.

Quantitative Evaluation: The results of our model and the comparison methods, based on no-reference metrics, are shown in Table IV. Our SCI++ model consistently ranked within the top positions across all metrics on the challenging in-the-wild dataset. In particular, SCI++ achieved lower LOE and NIQE scores. The improvements in these no-reference metrics indicate that SCI++ effectively handles diverse real-world conditions while preserving naturalness and minimizing distortions. Furthermore, SCI++ shows robustness across various challenging environments, outperforming other models in maintaining high visual quality without relying on reference images.

Qualitative Evaluation: As demonstrated in Fig. 10, through a large number of experiments, it can be seen that our method achieved more satisfactory visualization results than others, especially in the exposure level, structure depiction, color

[5]The newly-constructed dataset can be found in our project page.

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TABLE III
QUANTITATIVE RESULTS IN TERMS OF THREE FULL-REFERENCE METRICS (PSNR, SSIM, AND LPIPS), AND FOUR NO-REFERENCE METRICS (DE, EME, LOE, AND NIQE) ON THE MIT AND LSRW DATASETS

Method	MIT							LSRW						
	PSNR↑	SSIM↑	LPIPS↓	DE↑	EME↑	LOE↓	NIQE↓	PSNR↑	SSIM↑	LPIPS↓	DE↑	EME↑	LOE↓	NIQE↓
RetinexNet	14.120	0.747	0.281	6.692	13.467	1548.7	4.042	16.240	0.488	0.441	6.657	13.359	809.6	3.884
FIDE	17.560	0.791	0.210	6.771	13.650	224.7	3.980	18.148	0.646	**0.276**	6.830	8.832	188.4	4.405
KinD	17.549	0.854	0.180	7.013	12.209	457.0	3.341	18.302	0.633	0.290	6.855	9.459	441.3	3.616
DCC	21.262	0.870	0.167	7.107	10.744	238.5	3.323	19.526	**0.652**	0.300	**7.232**	9.428	223.0	4.113
URetinex	16.329	0.820	0.175	6.965	8.864	482.2	3.113	19.004	0.642	**0.259**	6.940	6.896	183.8	3.976
PairLIE	14.427	0.778	0.261	6.954	9.999	516.4	4.583	18.946	0.622	0.310	6.822	8.746	309.5	3.690
EnGAN	17.962	0.869	0.159	7.302	11.718	790.4	2.982	18.533	0.604	0.306	7.134	10.921	624.4	**3.242**
ZeroDCE	17.626	0.845	0.164	6.711	11.179	434.8	3.014	18.293	0.608	0.299	6.719	16.544	347.2	3.578
RUAS	19.198	0.912	0.153	**7.394**	15.550	676.4	2.959	15.828	0.648	0.382	7.052	11.570	487.2	4.993
CLIP-LIT	19.734	0.887	0.155	6.867	14.540	722.9	3.119	15.082	0.568	0.329	6.407	21.003	573.8	3.741
QuadPrior	20.566	0.857	0.176	7.141	10.873	585.6	3.648	18.929	**0.664**	0.334	7.016	6.695	550.1	5.210
ZERO-IG	11.371	0.723	0.278	7.240	42.037	420.3	3.810	18.483	0.531	0.359	**7.255**	**24.537**	381.2	3.669
SCI	**25.683**	**0.943**	**0.105**	7.268	**15.641**	**140.7**	**2.930**	**19.789**	0.611	0.284	6.993	20.804	**75.0**	3.616
SCI++	**25.826**	**0.948**	**0.108**	**7.399**	**15.751**	**141.5**	**2.924**	**19.835**	0.597	0.286	7.163	**21.193**	**137.8**	**3.493**

The best and second results are highlighted with red and blue, respectively.



Fig. 8. Visual comparison on the MIT dataset among state-of-the-art low-light image enhancement approaches.

TABLE IV
QUANTITATIVE RESULTS ON THE MIXED IN-THE-WILD DATASETS

Metrics	*Supervised Learning Methods*						*Unsupervised Learning Methods*							
	RetinexNet	FIDE	KinD	DCC	URetinex	PairLIE	EnGAN	ZeroDCE	RUAS	CLIP-LIT	QuadPrior	ZERO-IG	SCI	SCI++
DE↑	6.930	6.828	7.105	7.198	7.148	7.056	**7.273**	7.104	6.500	7.044	7.114	6.802	7.216	**7.379**
EME↑	14.421	11.423	10.343	9.869	11.191	9.386	11.733	14.834	12.257	18.094	8.042	**33.986**	16.856	**18.384**
LOE↓	713.5	215.6	256.5	199.1	232.0	304.3	684.2	273.2	961.2	498.8	423.136	495.899	**128.6**	**180.1**
NIQE↓	4.137	4.544	3.545	3.803	3.613	3.892	**3.463**	3.727	5.061	3.619	4.795	4.287	3.683	**3.518**

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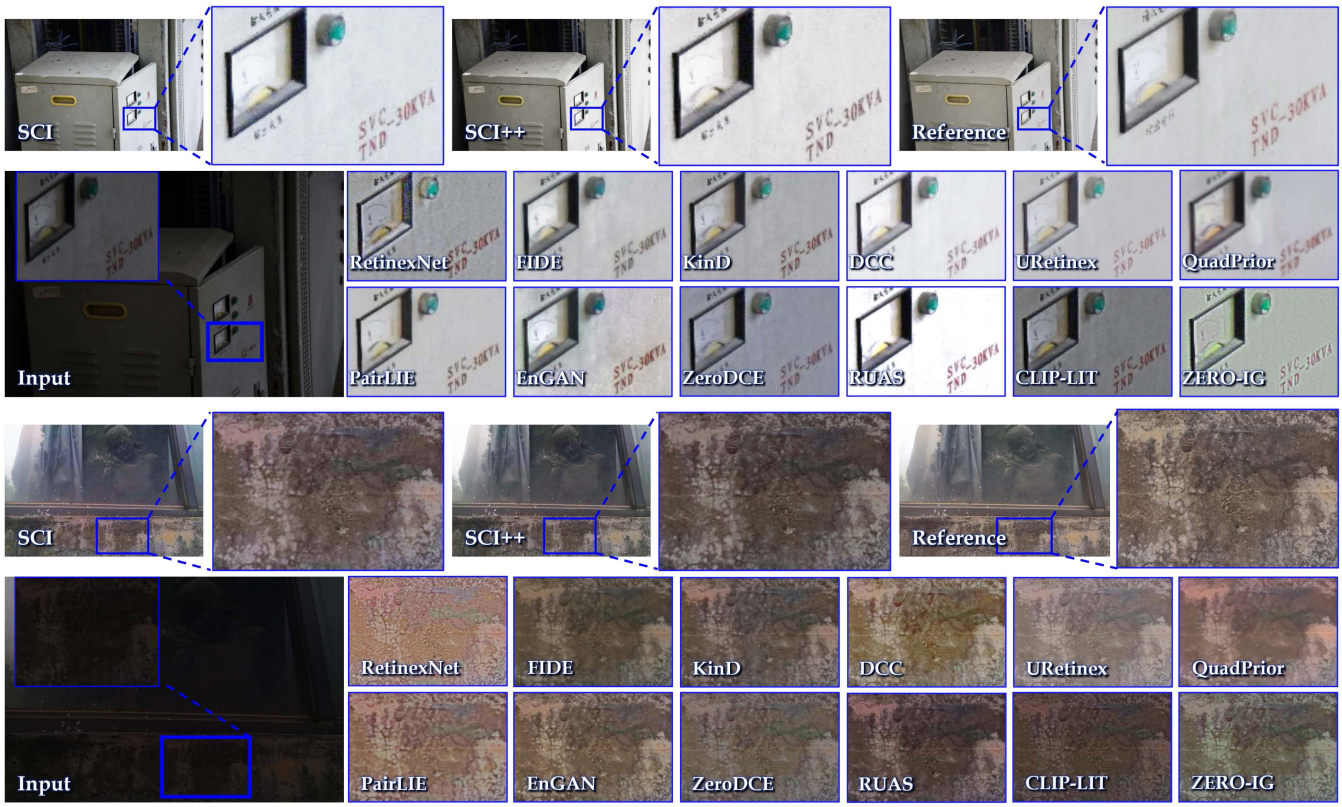


Fig. 9. Visual comparison on the LSRW dataset among state-of-the-art low-light image enhancement approaches.

TABLE V
THE MODEL SIZE, FLOPS (4K RESOLUTION) AND RUNNING TIME (GPU-SECONDS FOR INFERENCE) OF STATE-OF-THE-ART METHODS AND TWO VERSIONS OF OUR ALGORITHM

Metrics	RetinexNet†	FIDE†	KinD	DCC	URetinex	PairLIE	EnGAN	ZeroDCE	RUAS	CLIP-LIT	QuadPrior†	ZERO-IG	SCI/SCI++
SIZE (M)	0.8383	8.6213	8.5402	13.1887	0.3401	0.3418	8.6360	0.0789	**0.0014**	0.2788	1313.6	0.3302	**0.0003**
FLOPs (G)	-	-	2504.0	9377.6	7205.6	2828.3	2078.9	656.9	**27.1**	2304.1	-	4103.1	**4.4**
720P TIME (S)	0.6290	0.0793	0.7258	0.0065	0.0771	0.0014	0.0268	**0.0011**	0.0071	0.0019	3.926	0.0441	**0.0006**
1080P TIME (S)	1.4302	0.2048	1.6041	0.0074	0.1786	0.0017	0.0652	**0.0013**	0.0149	0.0022	-	0.0997	**0.0007**
4K TIME (S)	-	-	6.4136	0.0138	0.7219	0.0041	0.2162	**0.0019**	0.0697	0.0049	-	0.1784	**0.0007**

† denotes this work requires too much video memory on the high-resolution images, leading to a failure in producing results.

presentation, *e.g.,* in the first set of comparison images, only SCI and SCI++ produced clearer structures and accurate exposure, whereas other methods exhibited noticeable color casts, and in the second set of comparison images, the proposed method clearly demonstrated superior exposure control and detail recovery.

D. Computational Efficiency

Here we conducted a detailed analysis of the computational efficiency. Specifically, we reported the model size, FLOPs (Floating Point Operations), and running time (measured in GPU seconds) to provide a more comprehensive assessment. These results are summarized in Table V.

To ensure the comparison covered a range of practical use cases, we evaluated the methods using three standard image resolutions: 720P, 1080P, and 4 K, which are widely adopted in both consumer and industrial applications. Higher resolutions such as 4 K present significant challenges due to the increased computational demands, often leading to bottlenecks in processing time and memory usage. As our findings show, several recent CNN-based methods, including RetinexNet and FIDE, encountered hardware limitations at 4 K resolution. These models required substantial computational resources, which exceeded the available device memory or processing capacity, preventing accurate measurement of their computational costs and running times in such high-resolution scenarios.

In contrast, our proposed SCI model demonstrated significant advantages in terms of computational efficiency. SCI is the most lightweight model among the compared networks, exhibiting the lowest model size and FLOPs across all tested resolutions. This efficiency is particularly evident at 4 K resolution, where SCI continues to operate effectively, while other methods face considerable difficulties. This experiment highlights that SCI excels in terms of efficiency, providing distinct advantages over

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Fig. 10. Visual comparison on some in-the-wild challenging examples.

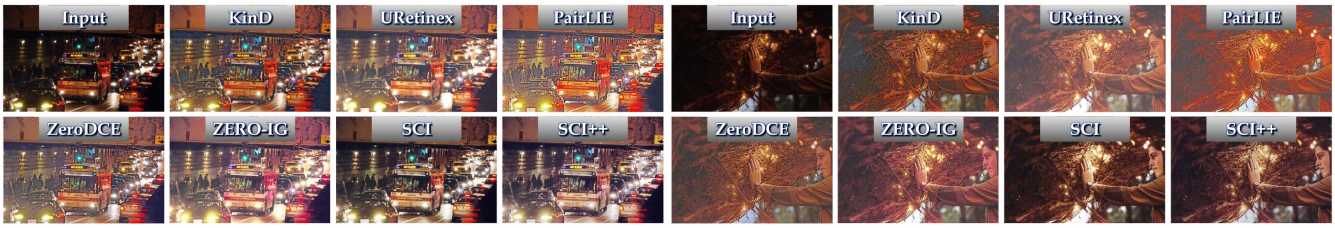


Fig. 11. Visual comparison on uneven illumination conditions. Zoom in for better view.

other state-of-the-art models, particularly in scenarios requiring high resolution or rapid processing.

E. Algorithmic Analyses

This part provides a detailed analysis to explore the characteristics of the proposed algorithm, including comparing the decomposed components, evaluating the impact of the self-calibrator, exploring its architectural variations, and examining the necessity of alternative learning strategies.

1) Evaluating Effects of Self-Calibrator: Here performed different learning strategy, as illustrated in Fig. 12. The naive strategy with only convolutions (see (b)) tended to result in overexposed images. This is because directly estimating illumination without any refinement can lead to an exaggerated increase in brightness across the image, especially in already well-lit regions. As a consequence, important details in these regions are often lost, and the resulting images appear washed out with diminished texture and structure.

To address the challenge of overexposure, the strategy with residual structure (see (c)) between the illumination and the input image significantly suppressed overexposure and prevented excessive brightness. However, it tended to struggle with fine detail recovery, leading to images that appear flat and lack the richness of texture and contrast necessary for high-quality visual results. After introducing a self-calibrator, our method (see (d)) ensured that brightness was evenly distributed without causing overexposure in already illuminated regions.

2) Exploring Architecture of Self-Calibrator: To further explore the impact of network structure on the learning process

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(a) Input (b) w/o residual+w/o self-calibrator (c) w/ residual+w/o self-calibrator (d) w/ residual+w/ self-calibrator

Fig. 12. Effects of self-calibrator.

TABLE VI
QUANTITATIVE RESULTS OF PARAMETER ANALYSIS

Method	Metric	Self-Calibrator Number			Channel			Batch Size			Patch Size		
		1	3	5	3	8	16	1	2	4	Full	256	128
SCI	PSNR↑	24.500	**25.683**	24.471	**25.683**	24.652	24.291	**25.683**	24.472	25.401	25.502	**25.683**	25.004
	SSIM↑	0.933	**0.943**	0.932	**0.943**	0.927	0.922	**0.943**	0.936	0.940	0.936	**0.943**	0.936
	LPIPS↓	0.111	**0.105**	0.111	**0.105**	0.113	0.114	0.105	0.116	**0.100**	**0.103**	0.105	0.110
SCI++	PSNR↑	25.720	**25.826**	25.158	**25.826**	24.496	25.748	**25.826**	25.612	24.763	25.797	**25.826**	25.471
	SSIM↑	0.944	**0.948**	**0.948**	**0.948**	0.922	0.939	**0.948**	0.941	0.926	0.946	**0.948**	0.941
	LPIPS↓	0.110	**0.108**	0.155	**0.108**	0.112	0.100	0.108	**0.097**	0.123	**0.098**	0.108	0.099

TABLE VII
QUANTITATIVE RESULTS IN TERMS OF DIFFERENT ARCHITECTURES OF SELF-CALIBRATOR ON THE IN-THE-WILD DATASET

Metrics	Conv	Attention	Self-Attention	ResConv
SIZE (M)	0.015	0.057	0.017	0.015
FLOPs (G)	23.221	25.000	26.922	23.221
DE↑	7.058	6.962	7.243	7.379
EME↑	18.254	18.589	17.966	18.384
LOE↓	216.274	230.831	170.652	180.1
NIQE↓	3.867	3.856	3.948	3.518

and the effectiveness of the self-calibrator, we experimented with different architectural variations. Specifically, we designed several network structure forms incorporating convolutional blocks [60], residual blocks [61], and attention modules [62]. These structural variations were systematically evaluated to understand how each component affects the learning process and the overall image enhancement results.

As shown in Table VII, the self-attention mechanism excelled in detail enhancement and exposure correction, but it significantly increased the model size and computational complexity. In contrast, the ResConv architecture demonstrated excellent performance in terms of naturalness and computational efficiency, with a smaller model size, making it suitable for resource-constrained scenarios. The attention mechanism had a slight advantage in contrast enhancement, but its exposure correction capability was weaker, as indicated by the highest LOE value. Overall, considering the trade-offs between performance and efficiency, we choose the ResConv architecture as our preferred structure.

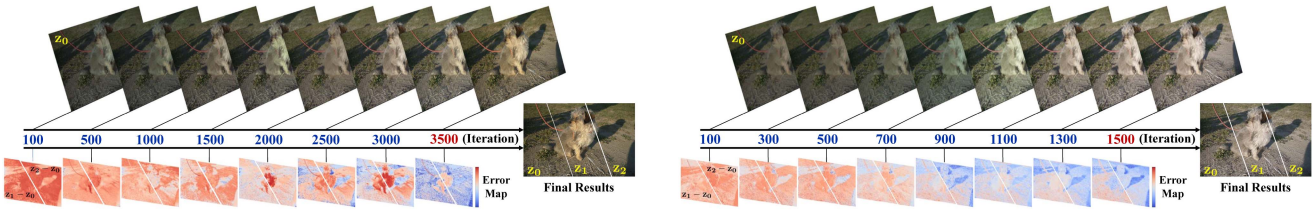
3) The Necessity of Alternative Learning: In the SCI framework, joint training of the enhancer and calibrator is essential for effective low-light image enhancement. However, in SCI++, an additive condition decouples the enhancer and calibrator, enabling more flexible and varied training strategies. This decoupling allows for independent training of the enhancer and calibrator, making applying different training configurations tailored to specific enhancement tasks possible. The learning process for each component can be more precisely controlled by alternating the training phases of the enhancer and calibrator rather than training them simultaneously. As shown in Table VIII, validation through experimental analysis had demonstrated that this alternating training approach results in improved enhancement performance compared to naive training.

TABLE VIII
THE NECESSITY OF ALTERNATIVE LEARNING

Model	PSNR↑	SSIM↑	DE↑	EME↑
Naive	25.079	0.939	7.243	17.568
Alternative	$25.826_{\uparrow 0.747}$	$0.948_{\uparrow 0.009}$	$7.379_{\uparrow 0.136}$	$18.384_{\uparrow 0.816}$

4) Comparing Decomposed Components: As shown in Fig. 14, we compared the estimated illumination maps generated by each model. Our SCI model stood out by producing illumination maps with remarkable smoothness. RetinexNet struggled with producing overly aggressive illumination maps that led to harsh lighting transitions, resulting in visible artifacts in the final enhanced result. While KinD was more refined than RetinexNet, occasionally failed to maintain consistency across areas with complex lighting conditions, which led to uneven enhancement in parts of the image. Although SSIENet was capable of generating relatively smooth illumination maps, tended to sacrifice some detail in the reflectance component, which impacts the overall sharpness and texture preservation. By contrast, the illumination map of our method maintained a consistent and gradual transition across the entire image, which translated to more visually pleasing effects.

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(a) The convergence behaviors of SCI (b) The convergence behaviors of SCI++

Fig. 13. Comparing convergence behaviors between SCI and SCI++. Zoom in for better view.

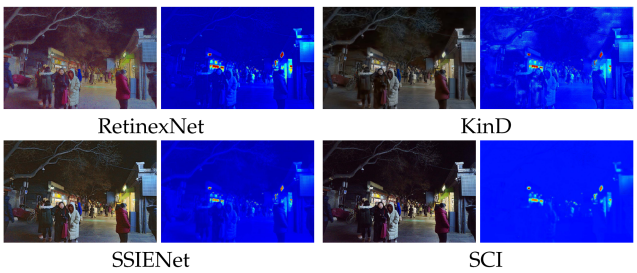


Fig. 14. Comparing the decomposed components among different illumination-based networks.

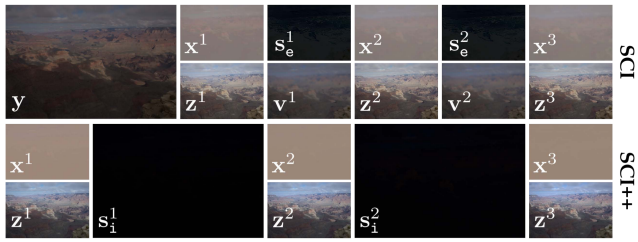


Fig. 15. Intermediate results of our proposed method.

5) Convergence Behaviors: As shown in Fig. 13, we compared the per-pixel error maps and reconstructed outputs at key iteration milestones for both SCI and SCI++. The baseline SCI exhibits slow error attenuation during early iterations, with error maps dominated by red halos around high-frequency edges. Even by 2000 iterations, residual structural artifacts persist as uneven error clusters, and only after approximately 3500 iterations does SCI achieve a predominantly small error distribution. In contrast, SCI++ achieves markedly faster and more stable convergence compared to the original SCI, owing primarily to the additivity assumption and the design of an independent self-calibrator. By invoking the additivity condition, it allows updates to be centered around a fixed initial estimate $\mathbf{x}^1$ rather than being continually reshaped by cascading dependencies.

6) Parameter Analysis: Our systematic parameter analysis on the MIT dataset (detailed in Table VI) evaluated four critical design variables: self-calibrator number, convolutional channel dimensions, batch size, and patch size. Empirical validation across both SCI and SCI++ architectures revealed a unified optimal configuration: three self-calibrator modules, three convolutional channels, a batch size of one, and a patch size of 256×256.

7) Intermediate Results: Fig. 15 shows the intermediate features of SCI and SCI++. In the original SCI, each stage alternates between illumination estimates $\mathbf{x}^t$, embedded self-calibrator outputs $\mathbf{s}_\mathrm{e}^t$, and $\mathbf{v}^t$. In the initial iterations, SCI already exhibited brightness bias and the illumination estimation results were not smooth, which affected the continuous illumination calibration in subsequent stages. By contrast, SCI++ begins with the same base illumination $\mathbf{x}^1$ and then applies independent self-calibrator maps $\mathbf{s}_\mathrm{e}^t$ directly to that fixed anchor. These maps $\mathbf{s}_\mathrm{e}^1, \mathbf{s}_\mathrm{e}^2$ remain sparse and focused, highlighting only the residual errors that require correction rather than re-estimating full illumination. This clear separation between a stable initial estimate and targeted residual refinements demonstrates how the additivity-driven independent self-calibrator in SCI++ mitigates error accumulation, yielding more interpretable intermediate maps and a smooth, monotonic improvement in visual fidelity.

VIII. Applications for Downstream Tasks

The low-light enhancer aims to improve visual quality to satisfy human eyes, and it also peels negative effects from degradation factors to some extent. In which, the unsupervised-type enhancer learns the latent regular defined by prior settings from natural images rather than learning specific distribution in the supervised pattern, thus it is intuitively helpful for downstream tasks. In addition, from the task's perspective, the low-light enhancer can be viewed as the data augmentation to provide task-friendly samples. A derivative fact is that the unsupervised low-light image enhancer can attain the task's applicability by pre-training it on task-specific data.[6] The remaining step is to fine-tune the task-specific network on the enhanced data. To explore it, we examine the role of the unsupervised low-light enhancer in following three types of generalized downstream tasks. In addition, for a more comprehensive comparison, we conduct additional comparisons with three representative supervised enhancers.

- *Pixel-Level Vision:* improving image quality under other degradations except for low-illumination. Here we considered image denoising [63], image super-resolution [64], and image deblurring [65].
- *Semantic-Level Vision:* extracting semantic information from the low-light observation. Dark face detection [66]

[6]The default pretraining iteration is set to 10^4.

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TABLE IX
QUANTITATIVE RESULTS ON PIXEL-LEVEL VISION TASKS

Method	*Denoising*			*Super-Resolution* (×4)			*Deblurring*		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Baseline	8.944	0.437	0.528	16.655	0.654	0.653	18.382	0.768	0.384
RetinexNet	21.208	0.885	0.247	16.431	0.739	0.757	15.257	0.523	0.504
URetinex	23.299	**0.941**	**0.118**	12.659	0.631	0.702	19.212	0.785	0.259
PairLIE	23.846	**0.935**	0.162	17.721	0.767	0.708	18.588	0.774	0.321
ZeroDCE	18.446	0.911	0.156	20.020	**0.781**	0.570	**25.169**	0.869	**0.168**
RUAS	20.064	0.899	0.233	19.989	**0.782**	**0.550**	17.367	0.618	0.598
ZERO-IG	23.852	0.933	0.170	19.803	0.774	0.560	20.100	0.824	0.227
SCI	**25.404**	0.932	0.135	**20.027**	0.780	**0.559**	25.046	**0.870**	0.169
SCI++	**25.513**	0.933	**0.133**	**20.241**	**0.781**	0.562	**25.193**	**0.872**	**0.167**



Fig. 16. Qualitative results of low-light image denoising.

and nighttime semantic segmentation [37] were performed as instances.

- *Other Vision:* reinforcing information representation in the other space except for the image space. Two typical tasks including novel view synthesis [67] and action recognition [68] were researched.

A. Pixel-Level Vision Tasks

We evaluate three pixel-level tasks under low-light conditions: denoising (LOL [8] dataset with frozen SCUNet [63]), super-resolution (RELLISUR [69] dataset using HAT [70]), and deblurring (LOL-Blur [65] dataset via FFTFormer [71]). The finetuning iterations of super-resolution and deblurring are both set to 10^4 Table IX summarizes performance improvements across three low-light vision tasks. Our method shows consistent advantages: 1) Denoising achieves competitive PSNR scores by better separating noise from image structures; 2) Super-Resolution obtains the most significant gains in both PSNR and SSIM through improved recovery of fine details in dark areas; 3) Deblurring enables more accurate edge restoration with enhanced contrast handling, reflected in sharper outputs. Figs. 16–17 visually demonstrate our method's advantages in noise suppression and detail preservation. Compared to RUAS and ZeroDCE, our approach maintains clearer edges in highlighted regions (red boxes) without over-smoothing artifacts, while achieving more accurate color reproduction in low-light textures (blue boxes). Specifically: 1) Text patterns and fine structures show improved legibility and continuity compared to the blurred outputs of other enhancers; 2) Shadow regions exhibit natural noise reduction without losing details.



Fig. 17. Qualitative results of low-light image super-resolution.

TABLE X
QUANTITATIVE RESULTS ON SEMANTIC-LEVEL AND OTHER VISION TASKS

Method	*Detection*	*Segmentation*	*View Synthesis*			*Action Recognition*	
	mAP(%)↑	mIoU(%)↑	PSNR↑	SSIM↑	LPIPS↓	Top-1(%)↑	Mean(%)↑
Baseline	0.596	0.461	6.842	0.150	0.583	0.175	0.152
ZeroDCE	0.645	0.452	17.390	**0.758**	0.472	0.189	0.161
RetinexNet	0.320	0.442	16.485	0.632	0.478	0.132	0.116
URetinex	0.523	0.455	15.624	0.704	0.425	0.162	0.141
PairLIE	0.468	0.444	15.872	0.678	0.465	0.094	0.082
RUAS	0.638	0.448	11.070	0.493	0.459	0.186	0.161
ZERO-IG	0.553	0.461	12.008	0.538	0.451	0.185	0.161
SCI	**0.663**	**0.463**	**19.928**	**0.775**	**0.438**	**0.194**	**0.166**
SCI++	**0.680**	**0.479**	**18.456**	0.751	**0.430**	**0.217**	**0.184**

B. Semantic-Level Vision Tasks

We evaluated two semantic-level vision tasks: dark face detection using DARK FACE [54] dataset with S3FD baseline [72] (metrics: accuracy/precision/recall, the finetuning iteration is set to 5×10^4), and nighttime semantic segmentation on ACDC [73] using PSPNet [74] (evaluated by mIoU, the finetuning iteration is set to 2×10^4). Table X benchmarks comparative performance on semantic-level tasks, where our SCI and SCI++ demonstrate dual-task superiority. SCI++ attains state-of-the-art metrics (mAP: 0.680↑, mIoU: 0.479↑), surpassing baselines by 9.8-21.3%. SCI maintains competitive performance with 0.663 mAP and 0.463 mIoU, validating architectural efficacy against both detection and segmentation baselines. Fig. 18 compares SCI/SCI++ with conventional enhancers on semantic tasks under low-light conditions. Zoomed regions reveal our methods' dual advantages: 1) Detecting finer objects (e.g., distant faces) missed by others in dark environments, 2) Generating artifact-free segmentation maps closely matching references. This visual superiority confirms enhanced capability in preserving structural fidelity and suppressing noise amplification during low-light semantic parsing.

C. Other Vision Tasks

We validated enhancers on two low-light vision tasks: novel view synthesis via LOM dataset [67] with NeRF baseline [75] (metrics aligned with pixel-level tasks, the finetuning iteration is

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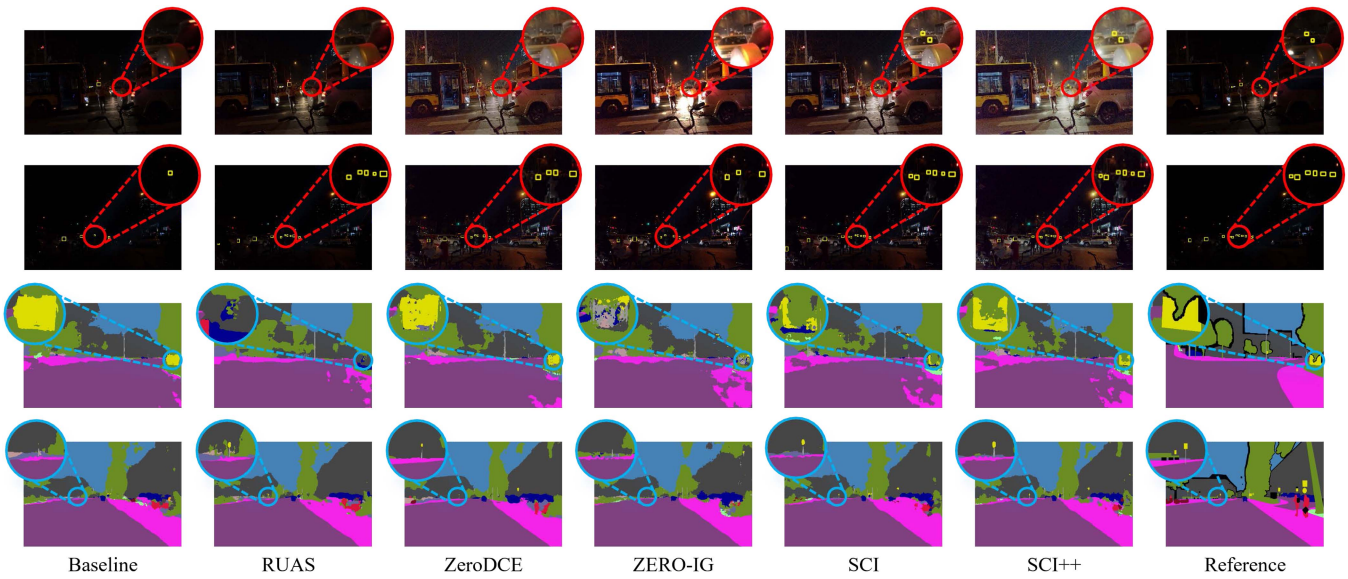


Fig. 18. Visual comparison results of dark face detection on the DARKFACE dataset and semantic segmentation on the ACDC dataset.

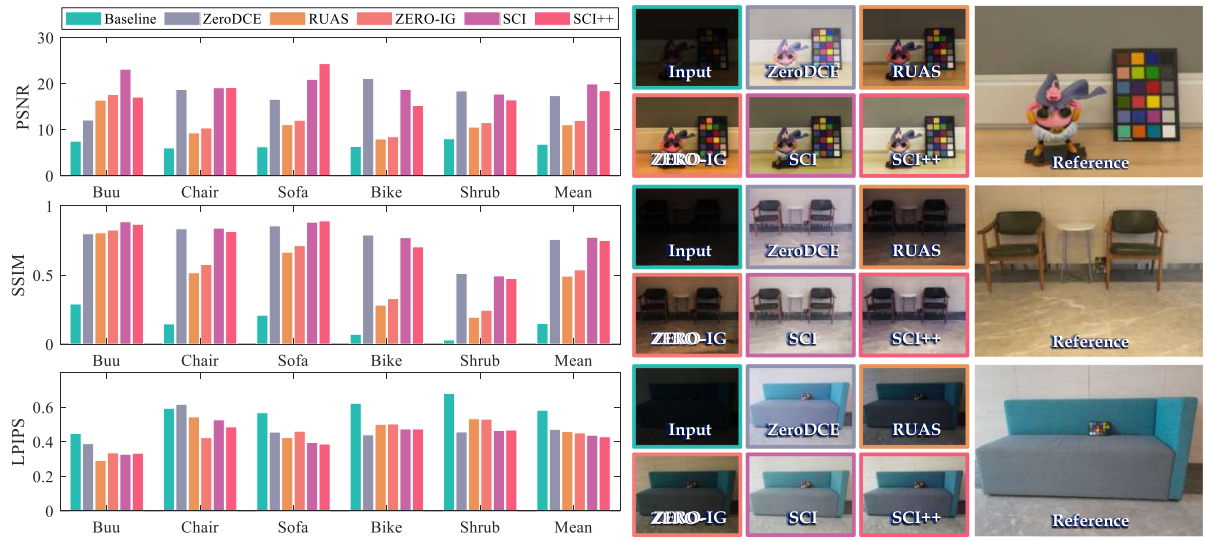


Fig. 19. Quantitative and visual results of new perspective generation on the LOM dataset [67].

set to 12.5×10^4), and action recognition using ARID [76] with frozen UniFormerV2 [77] (evaluated by Top-1/mean accuracy). SCI++ demonstrates cross-task superiority with state-of-the-art results: 1) Novel view synthesis achieves 19.928 PSNR (structural fidelity) and 0.775 SSIM (textural consistency), coupled with best perceptual quality (0.438 LPIPS); 2) Action recognition attains 21.7% Top-1 accuracy (+15.6% vs. baselines) and 18.4% mean accuracy, confirming robust feature preservation in low-light scenarios. Fig. 19 reveals critical advantages of SCI++ in novel view synthesis. While baseline methods (ZeroDCE/RUAS) suffer from underexposure-induced structural degradation, particularly losing Sofa textures and Buu figurine details, our method maintains photometric consistency with sharp edge preservation.

In summary, the experimental results across various downstream tasks demonstrated the significant advantages of incorporating our proposed SCI. By improving the quality of low-light inputs, it effectively enhanced both pixel-level and semantic-level tasks, allowing downstream models to achieve better performance in terms of accuracy, structural fidelity, and perceptual quality. For novel view synthesis, SCI provided clearer, more detailed reconstructions with improved geometric accuracy, while for action recognition in the dark, the enhanced frames facilitated better motion tracking and classification, resulting in higher recognition accuracy. These results confirm that our method not only addresses the challenges posed by low-light conditions but also strengthens the overall performance in a variety of complex downstream tasks.

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IX. Concluding Remarks

In this work, we proposed Self-Calibrated Illumination (SCI) learning scheme to successfully address the challenge of balancing quality and efficiency in low-light image enhancement. By introducing a weight-sharing illumination estimation process and a self-calibrator, the method significantly reduced computational costs while maintaining high performance. The reinforced SCI++ further disentangled the key components of the model, offering a more interpretable and effective learning framework. Experimental results demonstrated that the proposed approach not only improves image restoration quality but also enhances performance across various vision tasks, outperforming existing methods in both benchmarks and real-world applications.

Future research will explore hardware-efficient implementations of the self-calibration strategy through optimized convolutional operators while investigating temporal extensions to address dynamic feature interdependencies in video enhancement scenarios. These efforts aim to enhance both computational adaptability and sequential data processing capabilities.

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